```
print("Hello world")
print('5*10')
print(10+5)
```

```
Hello world
5*10
15
```

```
# divide in java int / int -> int
print(10/2)
print(10//2)
print(10 ** 2)
print("hello"+"world")
print("hi" * 5)

"""
bsdjfn
sdfhlksdjhf
vnskjdfhklsd
"""
```

```
5.0
5
100
helloworld
hihihihihi
```

```
a=5
print(type(a))
print(type("hdsjkfhskdjfhs"))
print(type(567.34534))
```

```
<class 'int'>
<class 'str'>
<class 'float'>
```

```
# variable names

A-Z, a-z, 0-9, _


total-score -> no
@value -> no
2023sfuigh -> no
```

```
for _ in rnage()
```

```
a=123
print(str(a))
c=True
d=False # bool
print(type(str(d)))
```

```
123
<class 'str'>
```

```
a=input() # user input
print(a)
print(type(a))
```

```
5
5
<class 'str'>
```

```
a=int(input()) # user input
print(a)
print(type(a))
```

```
5
5
<class 'int'>
```

```
print(10,20,30, end="")
print(10,20,30, sep="@")
# sep -> " "
# end -> "\n"
```

```
10 20 3010@20@30
```

```
# logical operator
'''
&& -> and
|| -> or
!= not
'''
```

```
print((2<3) and (2<4))
print((2>3) or (2<4))


# shortcircuit

# True and True -> True
# False and True second exp is never evaluated ->False


# False or True -> True
# True or ____ -> second exp is never evaluated -> True
```

```
True
True
```

```
# conditional

if, else, else if


if, else
elif
```

```
if 2<3:
  print("inside the if block")
else:
  print(" im in else")
```

```
inside the if block
```

```
if 2>3:
  print("inside the if block")
else:
  print(" im in else")
```

```
 im in else
```

```
if 2>3:
  print("inside the if block")

print("over here")
else:
  print(" im in else")
```

```
  File "/tmp/ipython-input-4041505273.py", line 5
    else:
    ^
SyntaxError: invalid syntax
```

```
if 2<3: # True
  print("if 1")
if 4>6: # False
  print("hdsjkhf")
else:
  print("hsdu")
```

```
if 1
hsdu
```

```
if 2<3: # True
  print("if 1")
elif 4>6:
  print("hdsjkhf")
else:
  print("hsdu")
```

```
if 1
```

```
# Fizz - Buzz
a%3==0 -> Fizz
a%5==0 -> Buzz
if both Fizz-Buzz
```

```
a = int(input())

if (a%3==0 and a%5==0):
  print("Fizz-Buzz")
elif a%3==0:
  print("Fizz")
elif a%5==0:
  print("Buzz")
```

```
15
Fizz-Buzz
```

```
# loops

# while
# initialize
# condition
# statement
# update

count=1
while(count<=10):
  print(count)
  count+=1
print("end")
```

```
1
2
3
4
5
6
7
8
9
10
end
```

```
# range
# range(start,end,jump)
# start=0 (inclusive)
# jump=1
# end is exclusive

range(5) # 0,1,2,3,4
range(1,6) # 1,2,3,4,5
range(1,10,2) # 1,3,5,7,9
range(-1,-5) # start=-1, end=-4, jump = +1 nothing
range(-1,-5,-1) # -1, -2, -3,-4
print(range(5,1)) # nothing
```

```
range(5, 1)
```

```
for i in range(5): # 0 -> 1 -> 2-> 3-> 4
  print(i)
```

```
0
1
2
3
4
```

```
for i in range(1,5,2):
  print(i)
```

```
1
3
```

```
for _ in range(5):
  print("a")
```

```
a
a
a
a
a
```

```
for i in range(5): # 5 rows
  for j in range(5): # 5 elements
    print("*",end=" ")
  print()
```

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

```
# functions

def is_valid(s):
  print("inside the function")


is_valid(4)
```

```
inside the function
```

```
# list -> []

# dynamic
# hetero
# stores objects

nums=[1,2,3]
mixed=[1,2.3,True,"1",'hsjkdhf']
empty=[]
```

```
colors=["red","black","yellow"]
print(colors)

first_ten=list(range(10))
print(first_ten)

chars=list("Section B")
print( chars)

print(colors[1])
print(colors[-1])
print(colors[-1][3])
colors.append("orange")
print(colors)

print("orange" in colors)
print("Orange" in colors)
```

```
['red', 'black', 'yellow']
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
['S', 'e', 'c', 't', 'i', 'o', 'n', ' ', 'B']
black
yellow
l
['red', 'black', 'yellow', 'orange']
True
False
```

```
for i in range(len(colors)):
  print(colors[i])
```

```
red
black
yellow
orange
```

```
for i in colors:
  print(i)
```

```
red
black
yellow
orange
```

```
num=list(range(100))
print(num)

# slicing
print(num[:10])
print(num[3:11])
print(num[1:11:2])
print(num[::-1])
```

```
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 2
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
[3, 4, 5, 6, 7, 8, 9, 10]
[1, 3, 5, 7, 9]
[99, 98, 97, 96, 95, 94, 93, 92, 91, 90, 89, 88, 87, 86, 85, 84, 83, 82, 81, 80, 79, 78, 77, 76, 75, 74, 73,
```

```
maths=[1,1,1]
sci=[2,2,2]
bio=[3,3,3]

subjects=[maths,sci,bio]
print(subjects)
print(subjects[1])
print(subjects[1][0])
```

```
[[1, 1, 1], [2, 2, 2], [3, 3, 3]]
[2, 2, 2]
2
```

```
# set
s=[1,2,3,4,4,4,4,4] #list
s={}
```

```
# dict
d={} # empty dict
print(type(d))

d={"a":64783, 2:"4", 4.5:True}
print(d)
print(d["a"])
print(d.get(2))
print(d.get(3))
print(d[3])
```

```
<class 'dict'>
{'a': 64783, 2: '4', 4.5: True}
64783
4
None
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
/tmp/ipython-input-3130663224.py in <cell line: 0>()
      8 print(d.get(2))
      9 print(d.get(3))
---> 10 print(d[3])

KeyError: 3
```

```
for key in d.keys():
  print(key)

print()
```

```
for key,value in d.items():
  print(key, value)


print()
for key in d.keys():
  print(d[key])
```

```
a
2
4.5

a 64783
2 4
4.5 True

64783
4
True
```

```
4 in d
```

```
False
```

```
s={1,2,3,3,1,1,2,2,3,4,4,5}
print(s)
```

```
{1, 2, 3, 4, 5}
```

```
import numpy as np

arr=np.array([10,20,30,40])
print(arr)

print(np.mean(arr))
print(np.median(arr))
print(np.var(arr))
print(np.std(arr))
print(np.min(arr))
print(np.max(arr))
print(np.sum(arr))
```

```
[10 20 30 40]
25.0
25.0
125.0
11.180339887498949
10
40
100
```

```
# np.random.random(size=None)
# decimal values 0,1
# returns float values
print(np.random.random(5))

#np.random.randint(low,high,size)
print(np.random.randint(1,7,10))

#np.random.choice(list,size)
print(np.random.choice(["Heads","Tails"],20))
```

```
[0.2003799  0.41235816 0.67655571 0.0979614  0.34403816]
[1 2 6 3 5 6 2 1 6 4]
['Tails' 'Tails' 'Tails' 'Tails' 'Tails' 'Heads' 'Tails' 'Heads' 'Tails'
 'Tails' 'Tails' 'Heads' 'Tails' 'Tails' 'Heads' 'Tails' 'Tails' 'Tails'
 'Heads' 'Tails']
```

```
# np.random.norma(loc,scale,size)

normal=np.random.normal(loc=50,scale=10,size=30)
print(normal)
```

```
[63.36128947 50.97820431 45.65999376 50.27525057 48.54190074 32.61757928
 46.08925654 69.75827836 46.36308277 57.21507155 55.77234457 58.51528925
 49.8372338  54.93252404 22.60299749 46.75667735 47.38381488 59.30239713
 70.92487862 60.29756341 60.1639303  40.31602498 49.54273062 48.95746385
 38.29583686 46.48651072 39.70389198 54.29135341 57.05042334 52.56359185]
```

Double-click (or enter) to edit